

# Baoli Hao

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🎓 [Baoli's Google Scholar](#)

## EDUCATION

### Illinois Institute of Technology

Aug. 2022 – Apr. 2027

*Ph.D. in Applied Mathematics*

Chicago, IL

- **Advisors:** Prof. Chun Liu & Prof. Ming Zhong
- **GPA:** 3.93/4.0
- **Research Interests:** Scientific Machine Learning (SciML) and Physics-Informed AI (e.g., PINNs, Operator learning); Inverse problems and system identification in ODE/PDE systems; Data-driven discovery and modeling of complex dynamical systems; Collective dynamics and large-scale interacting particle systems.

### City University of Hong Kong

Aug. 2017 – Apr. 2021

*BSc in Computing Mathematics (First Class Honours)*

Hong Kong

- **GPA:** 3.7/4.3 | **Minor:** Computer Science

## PUBLICATIONS

1. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. One Shot of Learning of Collective Dynamics from Equilibrium in Interacting Agent Systems. In preparation, 2026.
2. **Baoli Hao**, Kamrun Mily, Annalisa Quaini, and Ming Zhong. **A finite element framework for simulating residential burglary in realistic urban geometries.** *Mathematical Models and Methods in Applied Sciences*, 36(5), 1019–1050, 2026.
3. Kin Shing Chan, **Baoli Hao**, Kei Fong Lam, and Björn Stinner. **On a phase field model for binary mixtures of micropolar fluids with non-matched densities and moving contact lines.** *Interfaces and Free Boundaries*, accepted, 2025. arXiv:2504.21258.
4. **Baoli Hao**, Ulisses Braga-Neto, Chun Liu, Lifan Wang, and Ming Zhong. **Stability in training PINNs for stiff PDEs: Why initial conditions matter.** arXiv:2404.16189, 2024.
5. **Baoli Hao**, Ming Zhong, and Kevin O’Keeffe. **Attractive and repulsive interactions in the one-dimensional swarmalator model.** *Physical Review E*, 108(6), 064214, 2023.

## RESEARCH & PROJECTS

### Mathematical Modeling, Simulation & Analysis

- **Mathematical Modeling of Urban Crime Dynamics:** Developed and analyzed a nonlinear PDE system for crime hotspot formation derived from agent-based models via the mean-field limit. Implemented finite element simulations on realistic urban geometries, capturing spatial heterogeneity and natural boundary conditions. Published in *Mathematical Models and Methods in Applied Sciences (M3AS)* with open-source implementation available on [GitHub](#). Extended to a coupled multi-variable system incorporating adaptive intervention mechanisms, exhibiting feedback-driven suppression and emergent spatial oscillations. (Manuscript in preparation)
- **Ion Transport Models with Dynamic Boundary Conditions:** Developed and analyzed a novel Poisson–Nernst–Planck framework with cross-diffusion and dynamic boundary conditions for ion transport in electrochemical systems. Proved global existence of weak solutions and established stability properties, supported by numerical simulations capturing interfacial dynamics.
- **Emergent Behaviors in Synchronization and Swarming:** Developed and analyzed models of large-scale coupled oscillator systems with heterogeneous interactions, published in *Physical Review E*. Employed stability analysis and phase diagram characterization, combined with large-scale simulations, to classify emergent collective states. Current work extends to graph-structured networked systems.

- **Phase Field Modeling for Binary Micropolar Fluids:** Developed a thermodynamically consistent phase field model for binary mixtures of incompressible micropolar fluids, coupling a Navier–Stokes–Cahn–Hilliard system with micro-rotation dynamics. The model incorporates non-matched densities and moving contact line effects, and generalizes existing formulations. Established global existence of weak solutions in three dimensions for singular logarithmic and double-obstacle potentials.

## Inverse Problems & Data-Driven System Identification

- **Learning Collective Dynamics from Equilibrium Patterns:** Studied the inverse problem of recovering interaction laws in multi-agent systems from single steady-state observations. Addressed severe ill-posedness via distribution-based regularization, energy-based sign determination from the gradient flow structure, and a stopping-time strategy for time-scale recovery. Extended to multi-species and second-order systems.
- **Topological Interaction Learning in Multi-Agent Systems:** Developed a data-driven framework to learn interaction laws under adaptive and topological neighborhoods (e.g., KNN, radius-based, and density-weighted interactions), unifying a broad class of agent-based models. Enabled robust recovery of interaction kernels across deterministic and stochastic systems, including Langevin and self-propelled dynamics, with validation on both simulated and real-world data.

## Scientific Machine Learning & Physics-Informed AI

- **PINNs for Stiff Dynamical Systems:** Developed and benchmarked hard-constrained PINN architectures for stiff, time-dependent PDEs, achieving improved robustness, training stability, and accuracy over state-of-the-art methods across diverse benchmarks including sharp transitions and high-frequency coupled systems; grounded findings via Neural Tangent Kernel analysis and large-scale ablation studies.
- **Physics-Informed ML for Power System Stability:** Built an end-to-end pipeline for multi-bus power system transient stability, covering AC load flow initialization, fault scenario data generation, and a domain-decomposed energy-guided PINN in PyTorch with SIREN activations and Mixture-of-Experts (MoE) routing. Ongoing extensions target AI datacenter applications including burst event detection and grid stability analysis under high-variance GPU-driven load demands.

## TALKS & PRESENTATIONS

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|-------------------------------------------------------------------------------------------------------------|--------------|
| <b>SIAM Texas-Louisiana Sectional Meeting</b>                                                               | Sept. 2025   |
| Title: <i>Stability in training PINNs for stiff PDEs: Why initial conditions matter</i>                     | Austin, TX   |
| <b>Central AMS Sectional Meeting</b>                                                                        | Apr. 2025    |
| Title: <i>Modeling and analysis of Poisson–Nernst–Planck systems with steric effects and relative drags</i> | Lawrence, KS |
| <b>Summer School and Workshop on Computational &amp; Data Science at Duke</b>                               | Aug. 2023    |
| Title: <i>Mixed sign interactions in the 1D swarmalator model</i>                                           | Durham, NC   |

## INTERNSHIPS

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| <b>HKU Business School, The University of Hong Kong</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Jul. 2021 – May 2022 |
| <i>Data Scientist</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Hong Kong            |
| <ul style="list-style-type: none"> <li>• Supervisor: Dr. Qingchen Wang</li> <li>• Applied machine learning (LightGBM, logistic regression, NLP) and predictive analytics to large-scale electronic patient record (ePR) data covering approximately 90% of Hong Kong's population. Developed risk stratification and survival prediction models through feature engineering on clinical, laboratory, and prescription data. Utilized SHAP explainability to identify key predictors, improving model accuracy via cross-validation and hyperparameter optimization, and supporting data-driven healthcare decisions.</li> </ul> |                      |

## SERVICES

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|--------------------------------------------------------------------------------------------|------------------------|
| <i>Coordinator, Multiscale Modeling and Computation Seminar, Illinois Tech</i>             | Aug. 2022 – Present    |
| <i>Sponsorship and Student Support Chair, ACM HyperText 2025 Conference, Illinois Tech</i> | Oct. 2024 – Sept. 2025 |
| <i>Reviewer, AAAI Conference on Artificial Intelligence (AAAI 2025)</i>                    | Sept. 2025             |
| <i>Reviewer, ACM Transactions on Modeling and Computer Simulation (TOMACS)</i>             | Jan. 2026              |

## TEACHING

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### Illinois Institute of Technology

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|------------------------------------------------------------------------------------|---------------------|
| MATH Graduate Qualifying Exam Analysis Prep Session ( <i>Leader &amp; Mentor</i> ) | Fall 2025 – Present |
| MATH 500: Applied Analysis I ( <i>Teaching Assistant</i> )                         | Fall 2025           |
| MATH 578: Computational Mathematics II ( <i>Grader</i> )                           | Spring 2025         |
| MATH 151: Calculus I ( <i>Recitation Instructor</i> )                              | Spring 2023         |
| MATH 230: Introduction to Discrete Mathematics ( <i>Teaching Assistant</i> )       | Spring 2023         |
| MATH 252: Introduction to Differential Equations ( <i>Teaching Assistant</i> )     | Fall 2022           |

## HONORS & AWARDS

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| <b>Full Tuition Scholarship</b> , City University of Hong Kong | 2017 – 2021 |
| <b>Deans List</b> , City University of Hong Kong               | 2018 – 2021 |

## SKILLS

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**Programming** : Python, R, MATLAB, SQL

**Frameworks & Libraries**: PyTorch, JAX, TensorFlow, DeepXDE, FEniCS, Scikit-learn, SciPy, NumPy, Chebfun

**Languages** : English (Fluent), Mandarin (Native), Cantonese (Good)

**Others** : LaTeX, Microsoft Office, Git